

第十四章作业
14.4

$$V_o(s) = V_i(s) \cdot \frac{Z_L}{Z_{Total}} = V_i(s) \frac{Z_L}{Z_R + Z_L + Z_C}$$

代入可得 $V_o(s) = V_i(s) \frac{sL}{R + sL + \frac{1}{sC}}$, 则有

$$H(s) = \frac{V_o(s)}{V_i(s)} = \frac{sL}{R + sL + \frac{1}{sC}} = \frac{sL \cdot sC}{(R + sL + \frac{1}{sC})sC} = \frac{s^2 LC}{s^2 LC + sRC + 1}$$

同除以 LC , 得: $H(s) = \frac{s^2}{s^2 + \frac{R}{L}s + \frac{1}{LC}}$

14.30

(a) $f_0 = \frac{1}{2\pi\sqrt{LC}}$, 则 $C = \frac{1}{(2\pi f_0)^2 L}$, 代入得:

$$C = \frac{1}{(2\pi \cdot 15000)^2 \cdot 0.01} = \frac{1}{9 \times 10^6 \cdot \pi^2} = 1.127 \times 10^{-8} F$$

(b) 此时 $I = \frac{V}{R}$, 代入得 $I = \frac{120}{20} = 6 A$

(c) $Q = \frac{\omega_0 L}{R} = \frac{2\pi f_0 L}{R}$, 代入数值得 $Q = \frac{2\pi \cdot 15000 \cdot 0.01}{20} = \frac{300\pi}{20} = 15\pi = 47.12$

14.42

(a) $R = 2 + 6 = 8 \Omega$ $L = 1 H$ $C = 0.4 F$

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{1 \times 0.4}} = \frac{1}{\sqrt{0.4}} = 1.58 \text{ rad/s}$$

$$Q = \frac{\omega_0 L}{R} = \frac{1.58 \times 1}{8} = 0.198$$
 , $B = \frac{R}{L} = \frac{8}{1} = 8 \text{ rad/s}$

(b) $C = \frac{3 \times 6}{3 + 6} = 2 \mu F$, 则 $R = 2 k\Omega = 2000 \Omega$, $L = 20 \text{ mH} = 0.02 H$, $C = 2 \mu F = 2 \times 10^{-6} F$

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{0.02 \times 2 \times 10^{-6}}} = \frac{1}{\sqrt{4 \times 10^{-8}}} = 5000 \text{ rad/s}$$

$$Q = \omega_0 C R = 5000 \times 2 \times 10^{-6} \times 2000 = 20$$

$$B = \frac{\omega_0}{Q} = \frac{5000}{20} = 250 \text{ rad/s}$$

14.48

$$H(\omega) = \frac{R \parallel \frac{1}{j\omega C}}{j\omega L + R \parallel \frac{1}{j\omega C}} = \frac{\frac{R}{1 + j\omega RC}}{j\omega L + \frac{R}{1 + j\omega RC}} = \frac{R}{R + j\omega L - \omega^2 RLC}$$

$H(0) = 1$, $H(\infty) = 0$, 说明其为低通滤波器

14.55

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{0.05 \times 0.4 \times 10^{-6}}} = \frac{1}{\sqrt{10^{-8}}} = 10^4 \text{ rad/s}$$
 $f_0 = \frac{\omega_0}{2\pi} = 1591.5 \text{ Hz}$

$$B = \omega_2 - \omega_1 = \frac{R}{L} = \frac{10}{0.025} = 400 \text{ rad/s}$$
 , $Q = \frac{\omega_0}{B} = \frac{10000}{400} = 25$

由 $|\omega L - \frac{1}{\omega C}| = R$, 得 $\begin{cases} \omega_1 = \frac{-R + \sqrt{R^2 + \frac{4L}{C}}}{2L} \\ \omega_2 = \frac{R + \sqrt{R^2 + \frac{4L}{C}}}{2L} \end{cases}$, 代入得, $\begin{cases} \omega_1 = 9802 \text{ rad/s} \\ \omega_2 = 10202 \text{ rad/s} \end{cases}$

转换为频率: $\begin{cases} f_1 = \frac{9802}{2\pi} = 1560.1 \text{ Hz} \\ f_2 = \frac{10202}{2\pi} = 1623.8 \text{ Hz} \end{cases}$, 则 $1560.1 \text{ Hz} < f < 1623.8 \text{ Hz}$